UTS KASDAD *sisi 1 / 2, salin tulisan tangan*

0. Aturan main hitungan

- Bulatkan **di akhir**, 4 desimal. Isian SCeLE pakai titik. Nilai sangat kecil: tulis mantisa sesuai "$\times 10^{-k}$".
- Cek pembagi: kovarians/var $n-1$ (worksheet) vs N (slide [06] s7). Gaussian NB ws2: pembagi n. Ikuti soal.
- Kelas positif = kelas yang *penting* (ilegal, sakit, fraud).
- $\log_2 x = \ln x/0{,}6931$. $1/\sqrt{2\pi} = 0{,}3989$.
- Imbalanced: accuracy **dilarang**, stratified **wajib**, resampling/scaling/imputasi di-fit di **train** saja.
- Cek silang: $\sum \lambda$ = trace, kolom terpusat jumlah 0, posterior ternormalisasi jumlah 1, $|r| \le 1$, Gini anak $\le$ induk.

a. AI dan Data Science

	Humanly	**Rationally**
Thinking	cognitive modeling	laws of thought (logika)
Acting	Turing test	agen rasional

- Kritik: humanly = meniru manusia (pesawat $\neq$ burung); logika tak scalable; agen rasional paling bisa dimodelkan matematis.
- Agent function $f : \mathcal{P}^* \to \mathcal{A}$; agent program = implementasinya. Rasional = maks. *expected performance*.
- **PEAS**: Performance, Environment, Actuators, Sensors.
- Lingkungan: fully/partially observable, single/multi-agent, deterministic/stochastic, episodic/sequential, static/dynamic, discrete/continuous, known/unknown. **Taksi**: partial, multi, nondet, sequential, dynamic, continuous. Catur pakai jam: semi-dynamic.
- Agen: simple reflex (aturan kondisi-aksi) $\to$ model-based (state internal) $\to$ goal-based (search/planning) $\to$ utility-based (maks. expected utility); learning agent: performance element + learning element + critic.
- Supervised (label; y diskret = klasifikasi, kontinu = regresi), unsupervised (clustering), reinforcement (reward). ML = Representation + Evaluation + Optimization.
- Analytics: descriptive (apa), diagnostic (mengapa), predictive (akan apa), prescriptive (harus apa). Big Data 6V: volume, velocity, variety, veracity, value, variability.
- CRISP-DM: business und. $\leftrightarrow$ data und. $\to$ data prep $\leftrightarrow$ modeling $\to$ evaluation $\to$ deployment. KDD: selection $\to$ preprocessing $\to$ transformation $\to$ data mining $\to$ interpretation.

b. EDA dan data preparation

- Median: urutkan, posisi $(n+1)/2$; n genap rata-rata posisi $n/2$, $n/2{+}1$. Mode = frekuensi terbanyak. Range = (max $-$ min) (slide: +1 unit, contoh tanpa).
- **Kuartil** Q_1 = median separuh bawah, Q_3 = median separuh atas (median data *tidak ikut* bila n ganjil). $IQR = Q_3 - Q_1$.
- **Outlier** $RLB = Q_1 - 1{,}5\,IQR$, $RUB = Q_3 + 1{,}5\,IQR$. Boxplot: min, Q_1, median, Q_3, max.
- $s^2 = \frac{\sum(x-\bar{x})^2}{n-1}$. Skew $= \frac{\sum(x-\bar{x})^3}{ns^3}$: >0 ekor kanan (mean $>$ median). Kurtosis $= \frac{\sum(x-\bar{x})^4}{ns^4}$: normal 3, <3 platy, >3 lepto (outlier).
- Pearson $r = \frac{\sum(x-\bar{x})(y-\bar{y})}{\sqrt{\sum(x-\bar{x})^2\sum(y-\bar{y})^2}} \in [-1,1]$; $\chi^2 = \sum \frac{(O-E)^2}{E}$, $E = \frac{\text{baris}\times\text{kolom}}{N}$. Dipakai deteksi redundansi (integrasi).
- **Scaling** simple $x/x_{\max}$; min-max $\frac{x-\min}{\max-\min} \in [0,1]$; z-score $\frac{x-\mu}{\sigma}$. Wajib untuk KNN/PCA, tidak perlu untuk tree.
- **Missing** (1) tanya pengumpul data (2) drop baris/kolom (3) isi mean (numerik), mode (kategorikal), konstanta, nearest neighbor, fungsi/domain (4) biarkan. Muncul sebagai "?", N/A, 0, kosong.
- **Noisy** binning (smooth by mean/median/boundary), regresi, clustering, inspeksi manusia+komputer. Inkonsisten $\to$ data formatting.
- Integration: schema integration, entity identification, konflik nilai, redundansi (r, χ^2). Reduction: data cube, sampling, kompresi. Discretization: binning equal-frequency.

c. Feature engineering, selection, PCA

- Creation: domain, data-driven, aggregation, differences/ratios, age encoding (durasi hari), indicator (0/1 kondisi), binarization ($x > t \Rightarrow 1$), binning, BoW.
- Encoding: one-hot (nominal, kategori sedikit; jarak Manhattan antarkategori = 2); label/ordinal (berurutan); frequency (hitung kemunculan; dua kategori sama frekuensi tak terbedakan); target: regresi = mean y per kategori, klasifikasi = $P(y{=}1|k)$, berbobot $\frac{n\cdot\bar{y}_k + m\cdot\bar{y}}{n+m}$ (rawan leakage/overfit).
- Filter (cepat, sebelum model): missing ratio, variance threshold (normalisasi dulu), korelasi fitur-target ≈ 0 buang, antar fitur tinggi ambil satu, χ^2, IG. Wrapper: SFS (tambah terbaik), SBS (buang terburuk), SFFS (tambah lalu buang bersyarat), exhaustive, RFE. Embedded: tree, Lasso ($w = 0$).
- $H = -\sum p \log_2 p$; $IG(A) = H(S) - \sum_v \frac{|S_v|}{|S|} H(S_v)$.

$-p\log_2 p$: $\frac{1}{2}$ .5000 $\frac{1}{3}$ .5283 $\frac{2}{3}$ .3900 $\frac{1}{4}$ .5000 $\frac{3}{4}$ .3113 $\frac{1}{5}$ .4644 $\frac{2}{5}$ .5288 $\frac{3}{5}$ .4422 $\frac{4}{5}$ .2575 $\frac{1}{6}$ .4308 $\frac{5}{6}$ .2192 $\frac{1}{8}$ .3750 $\frac{3}{8}$ .5306 $\frac{5}{8}$ .4238 $\frac{7}{8}$ .1686. $H(\frac{1}{3},\frac{2}{3})$=.9183, $H(\frac{1}{4},\frac{3}{4})$=.8113, $H(\frac{2}{5},\frac{3}{5})$=.9710, $H(\frac{1}{5},\frac{4}{5})$=.7219, $H(\frac{3}{8},\frac{5}{8})$=.9544.

- **PCA** (1) $X_c = X - \mu$ (2) $S = \frac{X_c^T X_c}{n-1}$ (3) $\det(S - \lambda I) = 0$, $Sv = \lambda v$ (4) urutkan λ turun (5) $Z = T\,X_c^T$, T = eigenvector sebagai *baris* ($k \times p$), Z: $k \times n$.
- 2×2: $\lambda = \frac{tr \pm \sqrt{tr^2 - 4\det}}{2}$; $v \propto (b,\ \lambda - a)$ untuk $S = \begin{bmatrix} a & b \\ b & d \end{bmatrix}$, normalkan $\|v\| = 1$. Matriks blok-diagonal: eigen per blok; variabel tak berkorelasi $\Rightarrow \lambda$ = variansinya, v = sumbu itu.
- $EV_k = \lambda_k / \sum \lambda$; pilih k terkecil dengan kumulatif $\ge$ ambang.
- Rekonstruksi $\hat{x} = T^T z + \mu$; $SSE = (n-1)\sum \lambda_{\text{dibuang}}$; $MSE = SSE/(n\,p)$. Tanda eigenvector boleh dibalik (skor ikut berbalik).
- Curse of dimensionality (Hughes). t-SNE: proyeksi nonlinear untuk visualisasi (PCA dulu ke ± 50).

d. CART dan Random Forest

- Gini $= 1 - \sum p_c^2$; entropy $= -\sum p \log p$; pure $= 0$. Cost split $= \frac{|D_1|}{|D|}G_1 + \frac{|D_2|}{|D|}G_2$, pilih **min**.
- Numerik: urutkan, kandidat = midpoint ($n_{\text{unik}} - 1$ buah), uji $\le t$ vs $> t$. Kategorikal k nilai: partisi biner "$= t$ vs $\neq t$" (k buah).
- Urutan: Gini root $\to$ cost tiap fitur/threshold $\to$ root = cost min $\to$ anak pure jadi leaf $\to$ ulangi di anak impure $\to$ prediksi = mayoritas leaf.
- Regression tree: cost $= \frac{1}{|D|}\sum(y-\bar{y})^2$ dibobot; leaf = rata-rata y.
- Hyperparameter: max_depth, min_samples_split, min_samples_leaf, max_features, max_leaf_nodes. Parameter = hasil training (struktur, split).
- Plus: mudah dibaca, tanpa standardisasi, robust outlier, feature selection otomatis, surrogate split untuk missing. Minus: greedy, tidak stabil (**high variance**).
- Ensemble: regresi rata-rata, klasifikasi majority vote, stacking (bobot). Bagging: bootstrap N sampel *dengan pengembalian*; $\pm 63\%$ unik, $\pm 37\%$ out-of-bag (uji).
- RF: bootstrap + tiap **node** pilih $d \ll D$ fitur acak $\Rightarrow$ tree kurang berkorelasi. Hyper: n_estimators, max_samples (+ hyper tree). $d = D$ = bagging biasa.

e. Evaluasi model

aktual\pred	+	−
+	TP	FN
−	FP	TN

Acc $\frac{TP+TN}{\text{total}}$; Prec $\frac{TP}{TP+FP}$; Rec/TPR $\frac{TP}{TP+FN}$; Spec $\frac{TN}{TN+FP}$; FPR = 1−Spec.

- $F_1 = \frac{2PR}{P+R}$; $F_\beta = \frac{(1+\beta^2)PR}{\beta^2 P+R}$ ($\beta > 1$ recall, < 1 precision); G-mean = $\sqrt{\text{sens} \times \text{spec}}$; Brier = $\frac{1}{N}\sum(f_t - o_t)^2$.
- ROC: TPR vs FPR di semua threshold; AUC tak bergantung threshold. PR-AUC bila positif lebih penting. Threshold tinggi: FP turun, FN naik.
- Multikelas OvR. Macro = rata-rata metrik per kelas tanpa bobot. Micro = dari total TP, FP, FN (micro P = R = F1).
- MAE $\frac{1}{n}\sum|y-\hat{y}|$; MSE $\frac{1}{n}\sum(y-\hat{y})^2$; RMSE $\sqrt{MSE}$; $R^2 = 1 - \frac{\sum(y-\hat{y})^2}{\sum(y-\bar{y})^2}$ (negatif = sangat buruk). Residual harus acak.
- Holdout; train/val/test (val untuk tuning, test sekali); random subsampling; K-fold: uji N/K, latih $N(K-1)/K$, rata-rata error; LOOCV $K = N$ (data kecil). K besar: bias kecil, variance besar. Umum $K = 10$.
- Testing NOs: uji di data latih, tuning pakai test.
- Uji t: H_0, H_1, $\alpha \to S_p^2 = \frac{(n-1)S_1^2+(m-1)S_2^2}{n+m-2}$, $TS = \frac{\bar{X}_1-\bar{X}_2}{\sqrt{S_p^2(\frac{1}{n}+\frac{1}{m})}}$, $df = n+m-2 \to$ bandingkan dengan daerah tolak.

f. Bias-variance tradeoff

- $y = f(x) + \varepsilon$; $MSE = \text{Bias}^2 + \text{Var} + \sigma^2$ (σ^2 irreducible = batas bawah).
- Underfit: train & test error tinggi, dekat (high bias) → model lebih kompleks. Overfit: train sangat rendah, test tinggi (high variance) → pruning, regularisasi, ensemble, tambah data.
- Fleksibel ↑: bias ↓ var ↑. k-NN: $k \uparrow$ bias ↑ var ↓. Regularisasi: bias ↑ var ↓. Tree dangkal high bias, dalam high variance.
- Pre-pruning (kriteria heuristik: min sampel, max leaf, max depth) lebih cepat; post-pruning (tumbuh penuh lalu merge subtree).
- Bagging/RF paralel, turunkan var: $\text{Var}(\bar{y}) = \sigma^2/k$. Boosting serial, turunkan bias: AdaBoost (salah → bobot naik, resample, weighted vote), Gradient Boosting (awal = mean y, latih residual, jumlahkan $\alpha\times$output), XGBoost.

g. KNN

- Manhattan $\sum|p-q|$; Euclid $\sqrt{\sum(p-q)^2}$; cosine $\frac{p\cdot q}{\|p\|\|q\|}$; Jaccard $\frac{|p\cap q|}{|p\cup q|}$ (biner). Jarak kecil = dekat; similarity besar = dekat.
- Langkah: pilih $K \to$ jarak ke semua data latih → K terdekat → vote (klasifikasi) atau $\hat{y} = \frac{1}{K}\sum y$ (regresi). Tuliskan tabel jarak terurut; cek $K = 3$ vs $K = 5$.
- $K = 1$ rentan noise, K besar hilang lokalitas, pilih via CV. Wajib normalisasi; nominal one-hot. Tanpa training, testing $O(knd)$.

h. Imbalanced classification

- Slight 4:6, severe ≥1:100. Solusi: evaluasi (metrik, stratified), algoritma (cost-sensitive), data (over/undersampling).
- Bobot `balanced`: $w_c = \frac{n}{n_{\text{kelas}}\cdot n_c}$. Cost awal = kebalikan rasio (1:100 ⇒ FN 100, FP 1). Total cost = $c_{FN}FN + c_{FP}FP$.
- Stratified: tiap fold memuat $1/K$ dari tiap kelas.
- SMOTE: $s = \lambda a + (1-\lambda)b$, $0 \le \lambda \le 1$ (atau $a + \text{gap}\cdot\text{diff}$). Cek titik: $\lambda = \frac{s-b}{a-b}$ harus **sama** di semua fitur dan di $[0, 1]$.
- Borderline-SMOTE (k tetangga): #maj = k noise; $k/2 \le$#maj$< k$ **borderline** (di-SMOTE); #maj $< k/2$ safe. SVM-varian pakai support vector. ADASYN: bobot $\propto \frac{\#\text{maj}}{\#\text{min}}$.
- Under-*keep*: NearMiss-1 (rata-rata jarak ke 3 minoritas terdekat terkecil), -2 (ke 3 terjauh terkecil), -3 (terdekat per minoritas); CNN (1-NN, yang salah masuk subset). Under-*delete*: Tomek ($NN(a) = b$, $NN(b) = a$, beda kelas, hapus mayoritas); ENN (3-NN salah klasifikasi dihapus). OSS = Tomek+CNN; NCR = CNN+ENN; SMOTE+Tomek, SMOTE+ENN.
- Sampling **hanya** di data training (imblearn Pipeline), bukan test.

i. Probabilitas, likelihood, Naive Bayes

- $P(A \cup B) = P(A) + P(B) - P(A \cap B)$; $P(A^c) = 1 - P(A)$; $P(A|B) = \frac{P(A,B)}{P(B)}$; chain $P(A, B, C) = P(C|A, B)P(B|A)P(A)$.
- Independen: $P(A, B) = P(A)P(B)$ ($\neq$ mutually exclusive: $P(A \cap B) = 0$). Bersyarat: $P(A, B|C) = P(A|C)P(B|C)$; keduanya *tidak* saling menyiratkan.
- Total: $P(A) = \sum_i P(A|C_i)P(C_i)$. Bayes: $P(C|A) = \frac{P(A|C)P(C)}{P(A)}$. PPV kecil bila prevalensi kecil.
- Bernoulli $p^x(1-p)^{1-x}$; Binomial $\binom{n}{x}p^x(1-p)^{n-x}$; Normal $f(x) = \frac{1}{\sigma\sqrt{2\pi}}e^{-\frac{(x-\mu)^2}{2\sigma^2}}$ (density bisa > 1).
- Likelihood $L(\theta|D) = \prod p(x_i|\theta)$ (data tetap, θ berubah); $\log L = \sum \log p$ (cegah underflow); MLE koin $\hat{\theta} = k/n$.
- NB (generative): $P(c|x) \propto P(c)\prod_d P(x_d|c)$, ambil argmax, $P(x)$ boleh diabaikan; normalisasi: bagi jumlah skor semua kelas.
- Training: prior N_c/N; kategorikal N_{dck}/N_c; numerik μ_{dc}, σ^2_{dc} per kelas. Laplace: $\frac{N_{dck}+1}{N_c+K}$ (K = jumlah kategori fitur), terapkan ke **semua** kelas.
- Zero-frequency ⇒ posterior 0. Missing di data uji: abaikan sukunya. Campuran: $P(X|y) = P(X_n|y)P(X_c|y)$. Varian: Gaussian, Multinomial (hitungan), Bernoulli (biner), Categorical.

Urutan isian soal berantai (worksheet)

- **PCA**: mean tiap var → X_c (cek jumlah 0) → var, cov $(n-1)$ → $\det(S - \lambda I)$ per blok → λ urut → v ($\|v\| = 1$, tanda sesuai soal) → $\sum\lambda$, EV, kumulatif, $k \to$ ukuran T, $Z \to$ skor $z = v \cdot x' \to SSE$, MSE.
- **CART**: hitung kelas → Gini root → tiap fitur: Gini anak, cost → kategorikal: k partisi → numerik: midpoint, cost → root = min → leaf pure → split anak impure → prediksi data baru.
- **Metrik dari narasi**: tentukan kelas positif dulu → isi TP, FN, FP, TN (cek jumlah = total) → metrik.
- **Bayes test**: $P(S)$, sens = $P(+|S)$, spec = $P(-|\neg S) \to P(+)$ via total → $P(S|+)$; $P(S|-) = \frac{(1-\text{sens})P(S)}{1-P(+)}$.
- **Naive Bayes**: prior → $P(x_d|c)$ kategorikal (Laplace bila ada 0, ke semua kelas) → μ, σ^2 per kelas → $f(x|c)$ → skor = $P(c)\prod$ → normalisasi → argmax.
- **KNN**: tabel jarak semua titik, urutkan, tandai kelas → vote $K = 3$, $K = 5$ → regresi rata-rata.